INCH-POUND
MS27391F
8 May 2013
SUPERSEDING
MS27391E
5 February 2009

DETAIL SPECIFICATION SHEET

# NIPPLE, FLARELESS, TUBE TO HOSE - SWIVEL NUT

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-27272.

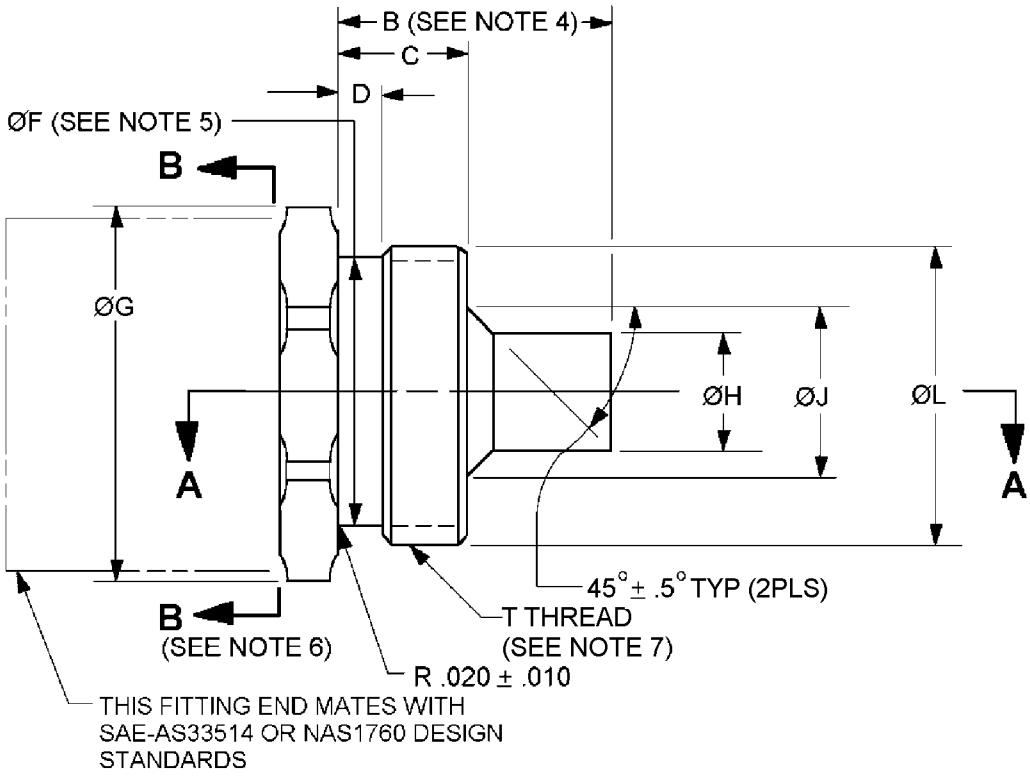
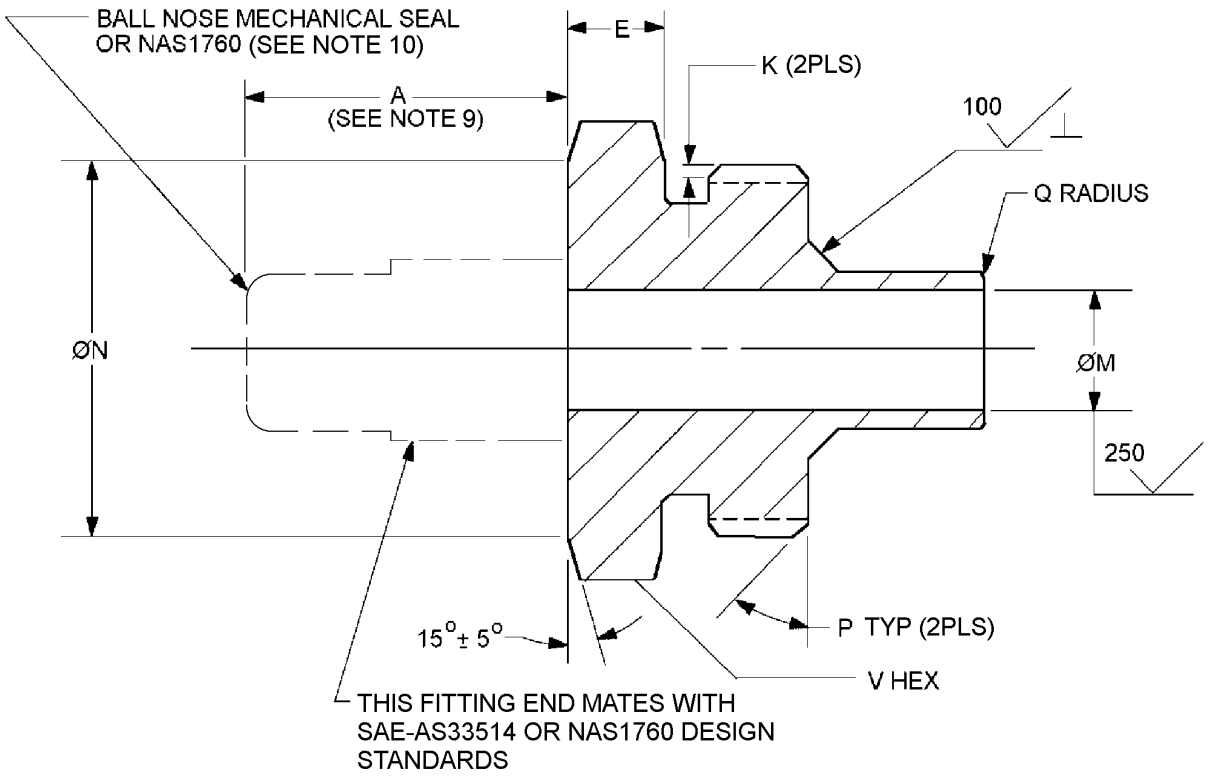



FIGURE 1. Nipple tube to hose dimensions and configuration.



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SECTION A-A ENLARGED

FIGURE 1. Nipple tube to hose dimensions and configuration - Continued.



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| Size and material code | | A min/max inches (mm) | B (see note 5) inches (mm) ±.005 (0.13) | | C inches (mm) ±.005 (0.13) | | D inches (mm) ±.005 (0.13) | |
|---|---|---|---|---|---|---|---|---|
| CRES | Al | | CRES | Al | CRES | Al | CRES | Al |
| -3/-4C | --- | .486/.543 (12.34/13.79) | .735 (18.67) | --- | .255 (6.48) | --- | .075 (1.91) | --- |
| -4C | --- | .440/.528 (11.18/13.41) | | --- | | --- | | --- |
| -5C | --- | .483/.598 (12.27/13.19) | .730 (18.54) | --- | .250 (6.35) | --- | .085 (2.16) | --- |
| -6C | --- | .558/.627 (14.17/15.93) | | --- | | --- | .050 (1.27) | --- |
| -8C | -8D | .662/.727 (16.81/18.47) | .870 (22.10) | .865 (21.97) | .270 (6.86) | .265 (6.73) | .085 (2.16) | .080 (2.03) |
| -10C | -10D | .755/.834 (19.18/21.18) | .950 (24.13) | .957 (24.31) | .300 (7.62) | .307 (7.80) | .100 (2.54) | .107 (2.72) |
| -12C | -12D | .733/.1.038 (18.62/26.37) | 1.035 (26.29) | 1.018 (25.86) | .360 (9.14) | .343 (8.71) | | .083 (2.11) |
| -16C | -16D | .819/.1.038 20.80/26.37) | 1.120 (28.45) | 1.100 (27.94) | .390 (9.91) | .370 (9.40) | .110 (2.79) | .090 (2.29) |
| -20C | -20D | .871/.1.120 22.12/28.45) | 1.335 (33.91) | 1.341 (34.06) | .400 (10.16) | .406 (10.31) | .085 (2.16) | .091 (2.31) |
| -24C | -24D | 1.096/1.377 (27.84/34.98) | 1.480 (37.59) | 1.480 (37.59) | .500 (12.70) | .500 (12.70) | .100 (2.54) | .100 (2.54) |

| Size and material code | | E inches (mm) ±.005 (0.13) | | F dia (see note 4) inches (mm) | | G dia inches (mm) | |
|---|---|---|---|---|---|---|---|
| CRES | Al | CRES | Al | | | | |
| -3/-4C | --- | .125 (3.18) | --- | .400 (10.16) | ±.005 | .620 (15.75) | ±.005 |
| -4C | --- | | --- | | | | |
| -5C | --- | | --- | .430 (10.92) | | .675 (17.15) | +.015 -.000 |
| -6C | --- | .135 (3.43) | --- | .495 (12.57) | | .750 (19.05) | |
| -8C | -8D | .150 (3.81) | .155 (3.94) | .600 (15.24) | | .960 (24.38) | |
| -10C | -10D | .170 (4.32) | .163 (4.14) | .785 (19.94) | ±.010 | 1.105 (28.07) | |
| -12C | -12D | .190 (4.83) | .207 (5.26) | .910 (23.11) | | 1.385 (35.18) | |
| -16C | -16D | .220 (5.59) | .240 (6.10) | 1.277 (32.44) | | 1.680 (42.67) | |
| -20C | -20D | .280 (7.11) | .274 (6.96) | 1.589 (40.36) | | 1.979 (50.27) | |
| -24C | -24D | | .280 (7.11) | 1.828 (46.43) | | 2.340 (59.44) | |

FIGURE 1. Nipple tube to hose dimensions and configuration - Continued.



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| Size and material code | | H dia inches (mm) +.000 -.005 (0.13) | J dia inches (mm) ±.005 (0.13) | K inches (mm) | |
|---|---|---|---|---|---|
| CRES | Al | | | | |
| -3/-4C | --- | .210 (5.33) | .295 (7.49) | .031 (.79) | ±.005 (0.13) |
| -4C | --- | | | | |
| -5C | --- | .273 (6.93) | .360 (9.14) | | |
| -6C | --- | .335 (8.51) | .425 (10.80) | .015 (.38) | max |
| -8C | -8D | .431 (10.95) | .530 (13.46) | .031 (.79) | ±.005 (0.13) |
| -10C | -10D | .531 (13.49) | .625 (15.88) | .035 (.89) | |
| -12C | -12D | .655 (16.64) | .760 (19.30) | | |
| -16C | -16D | .905 (22.99) | 1.040 (26.42) | .038 (.97) | |
| -20C | -20D | 1.156 (29.36) | 1.275 (32.39) | | |
| -24C | -24D | 1.406 (35.71) | 1.550 (39.37) | .043 (1.09) | |

| Size and material code | | L dia inches (mm) | | M dia inches (mm) | | N dia inches (mm) .02 (0.51) |
|---|---|---|---|---|---|---|
| CRES | Al | | | | | |
| -3/-4C | --- | .500 (12.70) | +.000/-.006 (+0.00/-0.15) | .161 (4.09) | +.005 (0.13) -.000 | .53 (13.46) |
| -4C | --- | | | | | |
| -5C | --- | .562 (14.27) | +.000/-.007 (+0.00/-0.18) | .224 (5.69) | | .59 (14.99) |
| -6C | --- | .625 (15.88) | | .261 (6.63) | | .66 (16.76) |
| -8C | -8D | .750 (19.05) | | .345 (8.76) | +.006 (0.15) -.000 | .84 (21.34) |
| -10C | -10D | .875 (22.23) | +.000/-.008 (+0.00/-0.20) | .440 (11.18) | | .97 (24.64) |
| -12C | -12D | 1.000 (25.40) | | .560 (14.22) | | 1.22 (30.99) |
| -16C | -16D | 1.375 (34.93) | +.000/-.009 (+0.00/-0.23) | .828 (21.03) | | 1.47 (37.34) |
| -20C | -20D | 1.688 (42.88) | | 1.058 (26.87) | | 1.78 (45.21) |
| -24C | -24D | 1.938 (49.23) | +.000/-.010 (+0.00/-0.25) | 1.282 (32.56) | +.005 (0.13) -.000 | 2.09 (53.09) |

| Size and material code | | P ±5º | | Q radius inches (mm) | |
|---|---|---|---|---|---|
| CRES | Al | CRES | Al | | |
| -3/-4C | --- | 45º | --- | .015 (0.38) | +.005 (0.13) -.000 |
| -4C | --- | | --- | | |
| -5C | --- | | --- | | |
| -6C | --- | 30° | --- | | |
| -8C | -8D | 45° | 30° | .020 (0.51) | ±.005 (0.13) |
| -10C | -10D | | | | |
| -12C | -12D | | | .030 (.76) | |
| -16C | -16D | | | | |
| -20C | -20D | | | | |
| -24C | -24D | | | .035 (0.89) | |

FIGURE 1. Nipple tube to hose dimensions and configuration - Continued.



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| Size and material code | | T (see note 7) | | V inches (mm) ±.02 (0.05) |
|---|---|---|---|---|
| CRES | Al | Thread | Pitch diameter Inches (mm) | |
| -3/-4C | --- | .5000 - 28 UNEF - 3A | .4768/.4740 (12.110/12.040) | .56 (14.22) |
| -4C | --- | | | |
| -5C | --- | .5625 - 24 UNEF -3A | .5354/.5325 (13.599/13.526) | .62 (15.75) |
| -6C | --- | .6250 - 24 UNEF -3A | .5979/.5949 (15.187/15.110) | .69 (17.53) |
| -8C | -8D | .7500 - 24 UNS -3A | .7229/.7198 (18.361/18.283) | .88 (22.35) |
| -10C | -10D | .8750 - 20 UNEF -3A | .8425/.8392 (21.401/21.316) | 1.00 (25.40) |
| -12C | -12D | 1.0000 - 20 UNEF -3A | .9675/.9641 (24.574/24.488) | 1.25 (31.75) |
| -16C | -16D | 1.3750 - 18 UNEF - 3A | 1.3389/1.3353 (34.008/33.917) | 1.50 (38.10) |
| -20C | -20D | 1.6875 - 18 UNEF - 3A | 1.6514/1.6476 (41.946/41.849) | 1.81 (45.97) |
| -24C | -24D | 1.9375 - 16 UN - 3A | 1.8969/1.8929 (48.181/48.080) | 2.12 (53.85) |

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, break or radius all corners .005, +.005, -.000 inches (-0.13, +0.13, -0.00 mm).
4. All diameters within B dimension shall be concentric within .005 inch (0.13 mm) full indicator movement. All other diameters must be concentric within .010 inches (0.25 mm) full indicator movement.
5. Thread gage must enter thread relief (sizes -10 through -24 only).
6. Any design to the left of plane B-B is acceptable provided the dimension A and the requirements of this specification sheet and the procurement specification are met.
7. Threads shall be in accordance with MIL-S-7742. Threads shall be rolled on corrosion-resistant steel only.
8. Dimension A indicates the overall length after mating parts are assembled. The measurement of the overall dimension "A" of the ghosted area is not measured to the end of MS27391, but is measured to the end of the mating part. It is not to the end of the ball nose mechanical seal of MS27391.
9. Unless otherwise specified, surface roughness shall not exceed 125 μin. $R_a$ in accordance with ASME B46.1.
10. If NAS1760 is used, MIL-DTL-25579/1 defines the method to adjust the hose assembly lengths to meet the original hose assembly lengths in MS8000 through MS8009.
11. Remove all burrs and slivers.

FIGURE 1. Nipple tube to hose dimensions and configuration - Continued.



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REQUIREMENTS

Nipple tube to hose dimensions and configuration, see figure 1.

This part is a integral part of MS27386. The item is only procured as an integral part of adapter assemblies.

Materials: Materials and material code see table I.

TABLE I. Materials and PIN codes.

| Material code | Material | Requirement |
|---|---|---|
| Blank | Combination of aluminum alloy and corrosion-resistant steel (CRES) | See below |
| C | CRES | Class 304, condition A, in accordance with ASTM A276 or 304 in accordance with SAE-AMS5639 |
| D | Aluminum alloy | 6061-T651, in accordance with SAE-AMS4117 |

Finish:

CRES: Passivate in accordance with SAE-AMS2700, method 1, type 6 or 7.

Aluminum alloy: Anodize in accordance with MIL-A-8625, type II, dye yellow.

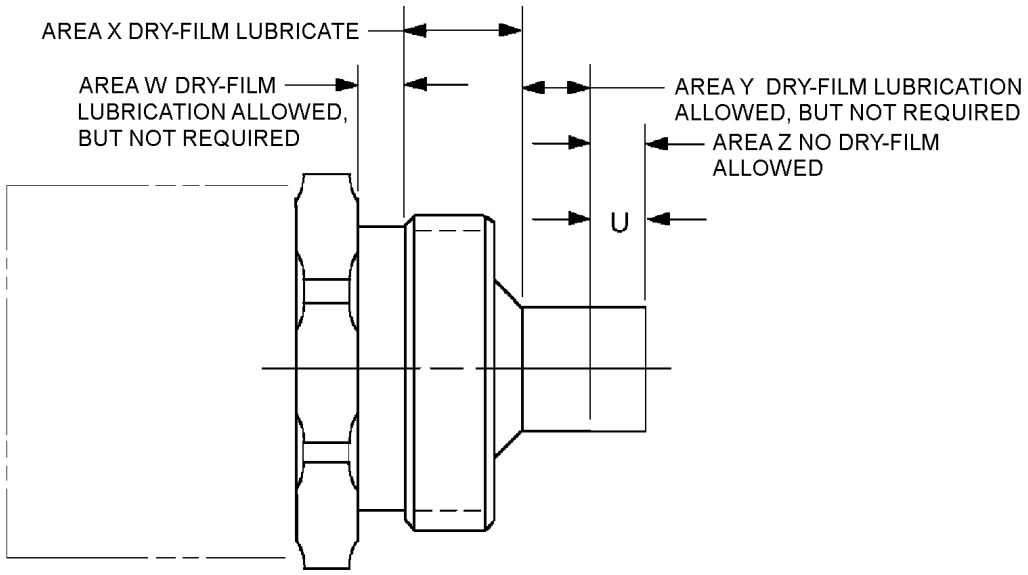
Lubrication:

NOTE: Avoid using graphite dry-film lubes with aluminum nipples because in a wet environment, graphite becomes corrosive to the aluminum.

Dry film lubricant. Dry-film lubricate CRES nipples in accordance with figure 2 and table II.



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| Size and material code CRES | U inches (mm) | |
|---|---|---|
| -3/-4C | .25 (6.35) | ±.05 (1.3) |
| -4C | | |
| -5C | | |
| -6C | | |
| -8C | .32 (8.13) | ±.12 (3.0) |
| -10C | .35 (8.89) | ±.15 (3.8) |
| -12C | | |
| -16C | .39 (9.91) | ±.19 (4.8) |
| -20C | .48 (12.19) | ±.28 (7.1) |
| -24C | .50 (12.70) | ±.30 (7.6) |

FIGURE 2. Dry-film lubrication area.



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TABLE II. Dry film designator.

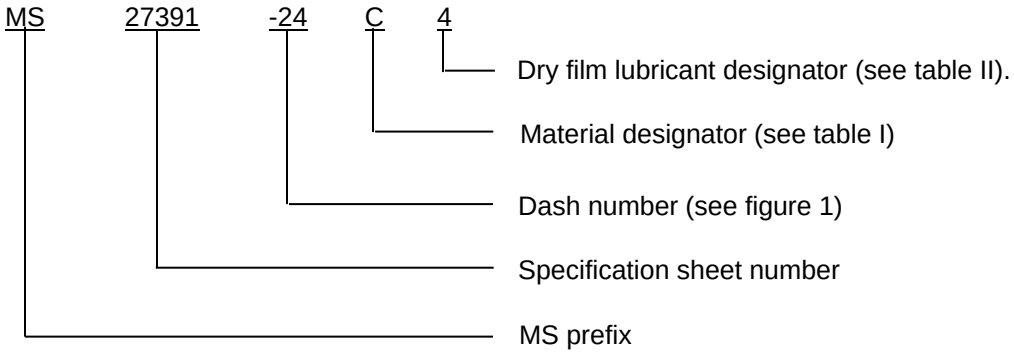
| Dry film designator | SAE class or type designator | Dry film characteristics |
|---|---|---|
| Blank | Any SAE class or type below | N/A |
| SAE-AS1701 | SAE-AS1701 class | SAE-AS1701 temperature ranges °F (°C) |
| 4 | 4 | -65° to +1400°F (-54° to 760°C) |
| 5 | 5 | -65° to +850°F (-54° to 454°C) |
| 6 | 6 | -375° to +850°F (-226° to 454°C) |
| SAE-AS5272 | SAE-AS5272 type | SAE-AS5272 temperature ranges. °F (°C) |
| 7 | Type I | -90° to 400°F (-68 to 204°C) endurance life of 250 min minimum |
| 8 | Type II | -90° to 400°F (-68° to 204°C) endurance life of 450 min minimum |
| 9 | Type III | Color 1 - Natural product color -90° to 400°F (-68 to 204°C) low Volatile organic compound with an endurance life of 450 min minimum |
| 10 | Type III | Color 2 - Black color -90° to 400°F (-68 to 204°C) low Volatile organic compound with an endurance life of 450 min minimum |
| Dry film designator | MIL classification | Dry film characteristics |
| MIL-PRF-46010 1/ | --- | MIL-PRF-46010 temperature ranges. °F (°C) |
| 11 | 1 | Color 1 natural product color, -90° to 400°F (-68 to 204°C) solvent resisting |
| 12 | 2 | Color 2 - Black color -90° to 400°F (-68 to 204°C) solvent resisting |

1/ Not for aerospace usage.



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Part or Identifying Number (PIN): The PIN shall consist of the prefix MS, specification sheet number, dash size number, material code letter, and solid film designator.




PIN example: MS27391-24C4 indicates a nipple hose to tube, size 1.500 inches, CRES, with dry film class designator 4.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-27272, this document references the following:

| | | |
|---|---|---|
| MIL-DTL-25579/1 | MS8004 | ASTM A276 |
| MIL-PRF-46010 | MS8005 | NAS1760 |
| MIL-A-8625 | MS8006 | SAE-AMS2700 |
| MIL-S-7742 | MS8007 | SAE-AMS4117 |
| MS8000 | MS8008 | SAE-AMS5639 |
| MS8001 | MS8009 | SAE-AS1701 |
| MS8002 | MS27386 | SAE-AS5272 |
| MS8003 | ASME B46.1 | SAE-AS33514 |



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## CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2013-034)

Review activities:
Army - AT, MI
Navy - MC, SA, SH
Air Force -71

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